Econ Theory (2012) 51:277–287
DOI 10.1007/s00199-009-0472-y

SYMPOSIUM

Multi-player contests with asymmetric information

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Received: 8 May 2008 / Accepted: 12 May 2009 / Published online: 28 May 2009
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Abstract We consider imperfectly discriminating, common-value, all-pay auctions (or contests) in which some players know the value of the prize, others do not. We show that if the prize is always of positive value, then all players are active in equilibrium. If the prize is of value zero with positive probability, then there is some threshold number of informed players such that if there are less, all uninformed players are active, and otherwise all uninformed players are inactive.

Keywords Contests · All-pay auctions · Asymmetric information

JEL Classification C72 · D44 · D72 · D82 · K41

1 Introduction

In a contest the players expend effort, resources, or money in order to increase their probability of winning a prize. Contest models therefore have important applications across the social sciences. Warfare, litigation, rent-seeking, campaigning in elections, and sports competitions are just a few examples of activities that have been modeled

I thank Patsy Haccou, Dan Kovenock, Johannes Münster, two anonymous referees, and participants at the 2005 Wissenschaftszentrum Berlin conference on "Advances in the Theory of Contests and Tournaments" and the 2008 CESifo Venice Summer Institute workshop on "Advances in the Theory of Contests and its Applications" for helpful remarks.

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as contests (see, e.g., Hirshleifer 1991, 1995; Skaperdas 1992; Bernardo et al. 2000; Tullock 1980; Baron 1989; Szymanski 2004). Studies of abstract contests include Dixit (1987), Gradstein and Konrad (1999), Myerson and Wärneryd (2006) and Wärneryd (2003). In this paper, we study the effects of introducing asymmetric information in a multi-player contest.

A contest may also be seen as an auction, specifically an *all-pay* auction, i.e., one where each participant must pay their bid. Hillman and Riley (1989) distinguish between *perfectly* and *imperfectly* discriminating all-pay auctions. In a perfectly discriminating auction one bidder wins with certainty when all bids are different. A well-known example of a perfectly discriminating all-pay auction is the war of attrition. While the properties of perfectly discriminating all-pay auctions have been studied extensively (see, e.g., Baye et al. 1993 or Krishna and Morgan 1997), much less is known about imperfectly discriminating models, where the bid profile induces a non-trivial probability distribution over the winner. From the standpoint of a seller offering a good in an auction a perfectly discriminating mechanism is optimal, but for applications such as those listed above imperfectly discriminating contests are frequently more realistic models.

We study common-value, imperfectly discriminating contests, i.e., contests where *ex post* all players would agree on the value of the prize, and a player who makes positive expenditure has a positive probability of winning. We further assume that some players may know the value of the prize with certainty at the point when they make their expenditure decision, others know only the prior distribution. We shall show, in particular, that if the prize is of positive value with probability one, then in equilibrium all players, informed as well as uninformed, spend positive amounts, and hence all have a positive probability of winning. This contrasts dramatically with equilibrium in a first-price auction in an otherwise similar setup. If there is more than one informed player in a common-value first-price auction, an uninformed player will not bid in equilibrium and will hence have a zero probability of winning (see, e.g., Milgrom and Weber 1982).

The existing literature on incomplete information in contests includes Hurley and Shogren (1998a,b) and Malueg and Yates (2004), who study models with independent valuations, Fu (2006), who studies endogenous timing of moves in a common-value contest with an informed and an uninformed player, Fey (2008), who considers asymmetric information about costs, and Münster (2009), who studies repeated games where abilities are private information. The present paper is an extension of the analysis of Wärneryd (2003) to the multi-player case.

In Sect. 2 we introduce a class of common-value contests with asymmetric information, and show that if the prize is positive with probability one, then all players are active in equilibrium. If there is a positive probability of the prize being of value zero, then there is some finite threshold number of informed players such that if it is met, uninformed players expend zero in equilibrium. Section 3 provides an example from the subclass of lottery contests, and shows that expected aggregate expenditure may be minimized at some mix of informed and uninformed players.

2 Multi-player contests

We consider $n \geq 2$ risk-neutral players participating in a contest for a prize of value y. The value y is distributed according to the continuous cumulative distribution function F with support on $[0, \hat{y}]$, with $\hat{y} > 0$. A number n_I of the players are privately informed about the value of the prize; the rest, numbering $n_U := n - n_I$, know only its prior distribution.

Player i spends effort or resources $x_i \in [0, \hat{y}]$ on increasing his probability

$$p_i(x_1, x_2, \ldots, x_n)$$

of winning the prize. At a most general level, we would only require that the p_i be non-negative, that they sum up to 1, and that a player's probability of winning be increasing in his own effort or expenditure and decreasing in everyone else's. Since we are ultimately interested in studying the effects of varying the numbers of players of different types, however, hence in effect comparing different games, we need a class of contests that meaningfully allows us to scale the size of the game up or down.

We shall therefore study contests from a class axiomatized by Skaperdas (1996). If player i spends x_i, then player i's probability of winning the prize is

$$p_i(x_1, x_2, \ldots, x_n) = \begin{cases} g(x_i)/\sum_j g(x_j) & \text{if } \sum_j g(x_j) > 0 \\ 1/n & \text{otherwise,} \end{cases}$$

where g is such that $g(0) = 0$, $g'(x) > 0$ for all x, and $g''(x) < 0$ for all x.

Consider now the optimal expenditure choice $x_i^I(y)$ of informed player i when he knows the value of the prize to be y. Define $G^I(y) := \sum_j g(x_j^I(y))$ and $G^U := \sum_j g(x_j^U)$. Assuming $G^U > 0$, the informed player i's expected utility is then

$$u_i^I(y) := \frac{g(x_i^I(y))}{G^I(y) + G^U} y - x_i^I(y),$$

which is concave in $x_i^I(y)$. Consider the first partial derivative of this utility with respect to own expenditure. It is

$$\frac{\partial u_i^I(y)}{\partial x_i^I(y)} = \frac{g'(x_i^I(y))(G_{\sim i}^I(y) + G^U)}{(G^I(y) + G^U)^2} y - 1. \tag{1}$$

This implies that all informed players expend the same amount in equilibrium. For suppose informed player i optimally expends the positive amount $x_i^I(y)$, i.e., that $\partial u_i^I(y)/\partial x_i^I(y) = 0$, and suppose there is some other informed player j who expends $x_j^I(y) \neq x_i^I(y)$. Without loss of generality, let $x_j^I(y) < x_i^I(y)$. We would then have that

$$\frac{\partial u_i^I(y)}{\partial x_i^I(y)} = \frac{g'(x_i^I(y))(g(x_j^I(y)) + G_{\sim i,j}^I(y) + G^U)}{(G^I(y) + G^U)^2} y - 1 = 0$$

and

$$\frac{\partial u_j^I(y)}{\partial x_j^I(y)} = \frac{g'(x_j^I(y))(g(x_i^I(y)) + G_{\sim i,j}^I(y) + G^U)}{(G^I(y) + G^U)^2} y - 1 \leq 0,$$

an impossibility since g is strictly concave. Similarly, if it is optimal for one informed player to be inactive, i.e., if $\partial u_i^I(y)/\partial x_i^I(y) \leq 0$ at $x_i^I(y) = 0$, then all other informed players are optimally inactive as well.

Hence all informed players expend zero if and only if we have that

$$\frac{g'(0)}{G^U} y - 1 \leq 0,$$

i.e., if

$$y \leq \frac{G^U}{g'(0)} =: y_0. \tag{2}$$

In particular, this implies $x^I(y) > 0$ for all $y > 0$ if we have $G^U = 0$.

Let $x^I(y)$ be the common equilibrium expenditure of informed players when the realized value of the prize is y. To summarize, we have that

$$\frac{g'(x^I(y))((n_I - 1)g(x^I(y)) + G^U)}{(n_I g(x^I(y)) + G^U)^2} y - 1 = 0 \quad \text{if } y > y_0 \tag{3}$$

and

$$x^I(y) = 0 \quad \text{if } y \leq y_0.$$

Setting the derivative in (1) equal to zero, differentiating with respect to y, and solving, we find that

$$\frac{\partial x_i^I(y)}{\partial y} = -\frac{g'(x_i^I(y))(G^I + G^U)}{y(g''(x_i^I(y))(G^I + G^U) - 2(g'(x_i^I(y)))^2)} > 0.$$

Hence each individual informed player's best-response function is nondecreasing in y, and the single-crossing property discussed in Athey (2001) is trivially satisfied.

Next consider uninformed player i, whose expected utility is

$$u_i^U := \int_0^{\hat{y}} \frac{g(x_i^U)}{G^I(y) + G^U} y \, \mathrm{d}F(y) - x_i^U,$$

which is concave in x_i^U. Note that since $G^I(y)$ is continuous and nondecreasing in y, the integral is well-defined. We have that

$$\frac{\partial u_i^U}{\partial x_i^U} = \int_0^{\hat{y}} \frac{g'(x_i^U)(G^I(y) + G_{\sim i}^U)}{(G^I(y) + G^U)^2} y \, \mathrm{d}F(y) - 1.$$

For reasons analogous to those in the case of the informed players, this implies that all uninformed players behave the same way when behaving optimally. Letting x^U be the common equilibrium expenditure of the uninformed players, this means that we have that

$$\int_0^{\hat{y}} \frac{g'(x^U)(G^I(y) + (n_U - 1)g(x^U))}{(G^I(y) + n_U g(x^U))^2} y \, \mathrm{d}F(y) - 1 = 0 \tag{4}$$

when $x^U > 0$, and

$$g'(0) \int_{y > y_0} \frac{1}{G^I(y)} y \, \mathrm{d}F(y) - 1 \leq 0 \tag{5}$$

when $x^U = 0$.

There cannot be an equilibrium such that the informed players are inactive, i.e., never expend anything. For suppose $x^I(y) = 0$ for all y. From (4) we would then have that

$$G^U = n_U g(x^U) = g'(x^U) \frac{n_U - 1}{n_U} \tilde{y},$$

and hence from (2) that

$$y_0 = \frac{g'(x^U)}{g'(0)} \frac{n_U - 1}{n_U} \tilde{y} < \tilde{y}$$

by the concavity of g. There would therefore be values of y occurring with positive probability such that at those values, the informed players would in fact want to make positive expenditure.

Consider now an equilibrium such that the uninformed players expend zero. We then have $y_0 = 0$, so the informed players make positive expenditure whenever y is positive. From (3) we have that

$$G^I(y) = g'(x^I(y)) \frac{n_I - 1}{n_I} y.$$

Substituting in (5), we have that

$$\int_{y > 0} \frac{g'(0)}{g'(x^I(y))} \frac{n_I}{n_I - 1} \, \mathrm{d}F(y) - 1 \leq 0. \tag{6}$$

Since we have $g'(0)/g'(x^I(y)) > 1$ by concavity of g, a necessary condition for this to hold is that $F(0) > 0$.

Because of the single-crossing property of the best-response functions of the informed players, the conditions for Corollary 2.1 of Athey (2001) are fulfilled, and a pure-strategy equilibrium exists. Hence we have the following.

Proposition 1 *There is a pure strategy equilibrium. If y only takes positive values, then in equilibrium all players are active.*

Next note that if the uninformed players are inactive, from (3) we have that

$$\frac{g(x^I(y))}{g'(x^I(y))} = \frac{n_I - 1}{n_I^2} y \quad \text{for all } y > 0. \tag{7}$$

Differentiating, we see that

$$\frac{\partial x^I(y)}{\partial n_I} = \frac{(n_I - 2) g(x^I(y)) g'(x^I(y))}{n_I(n_I - 1)(g(x^I(y)) g''(x^I(y)) - (g'(x^I(y)))^2)} < 0 \quad \text{for } n_I > 2.$$

Hence $x_I(y)$ is strictly decreasing in n_I for all y if we have $n_I > 2$. Furthermore, $x_I(y)$ must in fact converge to zero as n_I approaches infinity, since the right-hand side of (7) approaches zero. Hence for any y, both terms inside the integral in (6) approach 1 as n_I approaches infinity. If we have $F(0) > 0$, there must therefore be some finite $n_I^\star$ such that (6) holds for $n_I \geq n_I^\star$.

Furthermore, we have that

$$(1 - F(0)) \frac{n_I^\star}{n_I^\star - 1} - 1 < \int_{y>0} \frac{g'(0)}{g'(x^I(y))} \frac{n_I^\star}{n_I^\star - 1} \, \mathrm{d}F(y) - 1 \leq 0,$$

which implies that $n_I^\star \geq 1/F(0)$.

We have therefore proved the following.

Proposition 2 *Suppose we have $F(0) > 0$. Then there is some finite $n_I^\star$ such that if $n_I < n_I^\star$, then all players are active in equilibrium, but if $n_I \geq n_I^\star$, then only informed players are active in equilibrium. Furthermore, we have $n_I^\star \geq 1/F(0)$. Hence if we have $n < 1/F(0)$, all players are active in equilibrium.*

Intuitively, uninformed players are worse off the more informed players there are. This is because uninformed players face an analogue of the "winner's curse" problem of first-price auctions. Since there are better informed players in the game, an uninformed player cannot estimate the value of the prize at its prior expectation, but must discount it, taking into account that informed players expend less when the value is lower. The more informed players there are, the worse is the relative disadvantage.

The next section provides a simple example.

3 An example

3.1 Lottery contests

In this section we consider an example with $g(x) = x$.[1] This type of contest is sometimes known as a *lottery contest*. It was popularized by Tullock (1975); Tullock (1980) for the study of legal battles and rent seeking. For further discussion, see, e.g., Hirshleifer (1989). For an axiomatization, see Skaperdas (1996). Fullerton and McAfee (1999); Müller and Wärneryd (2001) and Nitzan (1991) is a small sample of the large body of literature on applications of this class of contests. Wärneryd (2003) exhaustively treats the two-player case with asymmetric information.

We now assume the prize is of value zero with probability $1-\pi$, and of value $\bar{y} > 0$ with probability π, with $\pi \in (0, 1)$. While this example is in some ways trivial, it has the great advantage of allowing for explicit analytical solutions. It also clearly shows the role of the number of informed players.

3.2 Case I: all players are active

In an equilibrium in which all players are active the first-order conditions imply that it must hold that

$$\frac{n_I x^I(\bar{y}) + (n_U - 1)x^U}{(n_I x^I(\bar{y}) + n_U x^U)^2}\pi\bar{y} = 1$$

and

$$\frac{(n_I - 1)x^I(\bar{y}) + n_U x^U}{(n_I x^I(\bar{y}) + n_U x^U)^2}\bar{y} = 1.$$

Solving this simultaneous equation system, in such an equilibrium we therefore have that

$$\begin{aligned} x^I(0) &= 0, \\ x^I(\bar{y}) &= \frac{(n-1)(\pi + (1-\pi)n_U)}{(n_U + \pi n_I)^2}\pi\bar{y}, \end{aligned}$$

and

$$x^U = \frac{(n-1)(1-(1-\pi)n_I)}{(n_U + \pi n_I)^2}\pi\bar{y}.$$

Uninformed players are therefore active if and only if we have $n_I < 1/(1-\pi) = n_I^\star$.

[1] Strictly speaking, this particular contest does not belong to the class studied previously, of course, since g is not strictly concave here.

This also allows us to conclude that if all players are informed, or all are uninformed, *ex ante* expected aggregate equilibrium expenditure is equal to $(n-1)\pi\bar{y}/n$. More generally, aggregate expenditure is

$$n_U x^U + \pi n_I x^I(\bar{y}).$$

3.3 Case II: only informed players are active

In an equilibrium where only informed players are active, which happens if and only if we have $n_I \geq 1/(1-\pi)$, we must have that

$$x^I(0) = 0$$

and

$$x^I(\bar{y}) = \frac{n_I - 1}{n_I^2}\bar{y}.$$

Expected aggregate equilibrium expenditure in such an equilibrium is therefore $\pi\bar{y}(n_I - 1)/n_I$.

3.4 Expected aggregate equilibrium expenditure

From an efficiency point of view we might be interested in how expected aggregate equilibrium expenditure $X := n_U x^U + \pi n_I x^I$ varies as a function of the relative shares of informed and uninformed players. Consider, therefore, X as a function of the number of informed players, keeping the total number of players, n, constant (throughout, we shall ignore that n, n_I, and n_U are necessarily integers).

Since the uninformed players will be inactive whenever we have $n_I \geq 1/(1-\pi) = n_I^\star$, and defining $D = n - n_I(1-\pi)$, we have that

$$X = \begin{cases} ((n-1)(n_I\pi^2 - (n-n_I)(n_I(1-\pi)^2 - 1))\pi\bar{y})/D^2 & \text{if } n_I < n_I^\star \\ \pi\bar{y}(n_I - 1)/n_I & \text{otherwise.} \end{cases}$$

Note that X is continuous in n_I at $n_I^\star$.

In case we have $\pi > (n-1)/n$, all players will be active in equilibrium, regardless of the relative numbers of informed and uninformed players, and in this case X is a strictly convex function of n_I, with a minimum at $n_I = n/(1+\pi)$. Figure 1 illustrates this case.

Next consider the case where we have $\pi \leq (n-1)/n$. Two subcases are of interest. It could be the case that we have $n_I^\star \geq n/(1+\pi)$, i.e., that $\pi \geq (n-1)/(n+1)$, in which case $n_I = n/(1+\pi)$ is still the number of informed players that minimizes expected aggregate equilibrium expenditure. Figure 2 illustrates this case. Note that the scales used in the figures are not commensurate—the values of the minimized expenditure in two figures may, of course, be different.

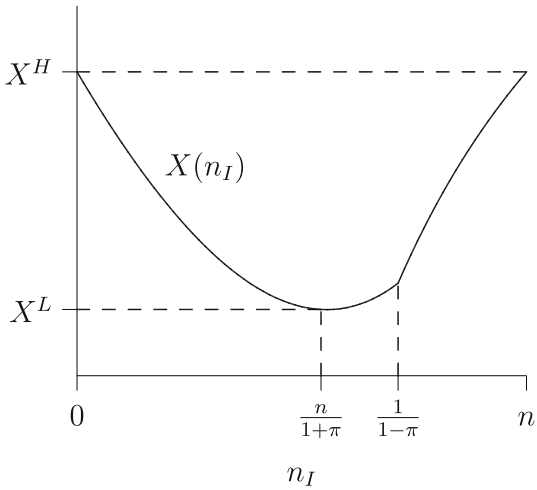

Fig. 1 Aggregate expenditure with $(n-1)/(n+1) \leq \pi \leq (n-1)/n$

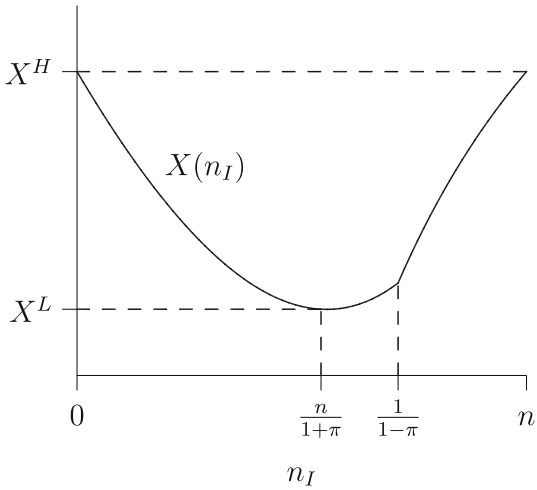

Fig. 2 Aggregate expenditure with $(n-1)/(n+1) \leq \pi \leq (n-1)/n$

Finally, it may be the case that $\pi < (n-1)/(n+1)$, in which case the minimum expenditure is reached at $n_I^\star$. This case is illustrated in Fig. 3.

In particular, we note that in each case that equilibrium expenditure is *minimized* with a population of players that includes both informed and uninformed players. Equilibrium expenditure is *maximized* when there are only players of one type.

4 Remarks

We have seen that in common-value contests with players who can be either perfectly informed of the value of the prize, or perfectly uninformed, both types of player will expend positively in equilibrium as long as the prize is always of positive value. In

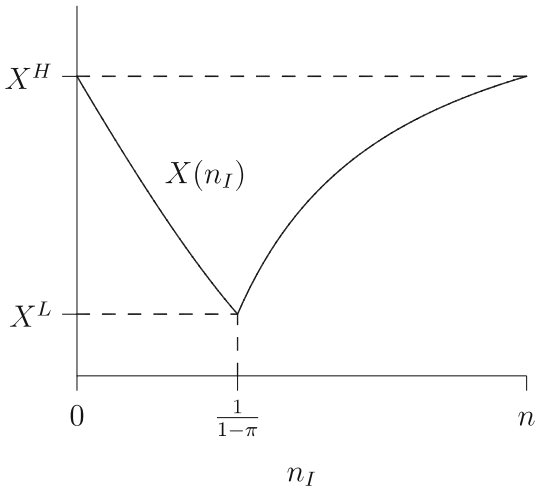

Fig. 3 Aggregate expenditure with $\pi \leq (n-1)/(n+1)$

the case of lottery contests we showed that there is a unique mix of informed and uninformed players that minimizes expected equilibrium expenditure.

The latter result suggests—in particular, if it can be generalized—that from the perspective of a contest designer who collects the expenditures of the contestants and wishes to maximize his revenue, having differentially informed contestants is never optimal. A risk neutral contest designer who does not himself know the value of the prize would prefer to have the contestants be either all informed or all uninformed. Since it seems reasonable to assume that the contest designer *does* typically know the value of the prize, however, the optimal composition of the population of contestants when the value is low is then for all of them to be uninformed—as they will, of course, typically spend more than would informed contestants. On the other hand, when the designer knows the value to be high, he would instead prefer to let the contestants know this. But such a strategy on the part of the designer would, of course, mean that revealing no information reliably signaled a low value. We leave the full analysis of this signaling game and its equilibria for future research.[2]

From the point of view of rent-seeking applications, in which the focus is often on the social waste of contest expenditure, we see that analyses that do not take asymmetric information into account may overestimate equilibrium rent dissipation. As in many applications it seems reasonable to assume that contestant are in fact asymmetrically informed, our results provide yet another clue to why empirical estimates of rent-seeking expenditures typically show them to be considerably lower than predicted by symmetric information models (see, e.g., Laband and Sophocleus 1992).

[2] I thank an anonymous referee for suggesting this exercise.

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